

NexusWallet

Adaptive Financial Intelligence

for the New Work Economy

An enterprise-grade, multi-agent GenAI platform that serves gig workers, creators, freelancers, independent contractors, and seasonal workers — the 40% of Southeast Asia's workforce that traditional banking still ignores.

TARGET TAM

\$87B underserved SEA workforce

CORE INNOVATION

Financial Digital Twin + RL Coach

STACK DEPTH

Multi-agent · RAG · RL · Digital Twin

GXS ALIGNMENT

Extends FlexiLoan + Pockets infra

01 · THE PROBLEM

A \$87B financial gap hiding in plain sight

GXS Bank has already proven there's a massive underserved market — **4 in 5 of its customers are underserved**, and FlexiLoan disbursed 200,000 loans in its first year. But GXS's current products treat all non-traditional workers the same. The reality is far more fragmented.

40%

of SEA's workforce is in non-traditional employment — gig, freelance, creator, seasonal

3×

more volatile monthly income for gig workers vs. salaried employees — traditional savings products fail them

\$169M

latest GXS Bank funding round (2024) — investor confidence in this exact segment exists, execution gap remains

"We're not competing for the same customer that the big bank is trying to get. There are so many people who are underserved that there is enough money to be made by serving them."

— Muthukrishnan Ramaswami, GXS Group CEO

⚠️ The Gap NexusWallet Solves

GXS FlexiLoan uses an "ecosystem risk score" based on Grab transaction data. This is powerful — but reactive. It tells you who a worker *has been*. NexusWallet builds a **Financial Digital Twin** that models who a worker *will be* across every earnings scenario, enabling truly proactive, personalized financial guidance — not just credit scoring.

Beyond gig workers: the full New Work Economy

The innovation here is NexusWallet's segment-aware intelligence. Each worker type has radically different income patterns, risk profiles, and financial anxieties. A one-size-fits-all coach is noise. NexusWallet builds *segment-specific financial personas* and adapts its guidance engine accordingly.



Gig Workers (Grab Drivers, Delivery Partners)

Daily cash-flow variability, peak/off-peak earnings, surge windows. Need micro-savings triggers, income smoothing, and accident/income-loss protection. **GXS ecosystem data is richest here** — NexusWallet becomes the intelligence layer on top of GXS FlexiLoan.

SEA GRAB PARTNERS: ~9M+



Creator Economy (YouTubers, Streamers, UGC Creators)

Royalty payment lag (30–60 day delays), multi-platform income aggregation, tax complexity. Creators in SEA earn on Shopee, TikTok, YouTube — none of these platforms offer financial planning. **First mover opportunity.** NexusWallet bridges Grab Pay + creator income streams.

SEA CREATOR ECONOMY: ~\$1.4B MARKET (2025)



Freelancers & Independent Contractors

Project-based income with multi-month dry spells, international client invoicing, unpredictable taxes. No employer-sponsored retirement. Often the highest-earning segment but worst at planning due to cognitive overload. The "brilliant but broke" problem.

SEA FREELANCERS: ~25M (2025 EST.)



Seasonal & Contract Workers

Tourism hospitality, harvest labor, festival economy workers. 3–6 month earnings windows followed by income cliffs. Critically underserved — no bank models their cyclical income. NexusWallet uses temporal modeling to plan across seasons, not just months.

SEA SEASONAL WORKFORCE: **~50M WORKERS**



Why This Segment Expansion is a Business Differentiator

GXS CEO has explicitly noted the need to "go outside of its established ecosystem to get enough volume." Creators, freelancers, and seasonal workers are *not currently Grab users* — but they will become GXS customers *through NexusWallet*, expanding the TAM beyond Grab's existing install base. This is a **new customer acquisition engine, not just a feature**.

03 · THE SOLUTION

NexusWallet: Three Intelligent Layers

NexusWallet is not a chatbot with financial tips. It is a **multi-agent, simulation-first financial intelligence platform** built on three deeply integrated layers. Each layer can deliver value independently, but their compound effect is what creates the moat.



Layer 1 · Core Innovation

The Financial Digital Twin (FDT)

A live, probabilistic simulation of each user's financial future. The FDT is not a dashboard — it is a **runnable model** that synthesizes income history, spending patterns, debt obligations, seasonal calendars, and macro signals (fuel prices for drivers, platform algorithm changes for creators, tourism cycles for seasonal workers) into a dynamic Markov model of the user's financial state.

What it models:

Income distribution per week/month (not just average), expense volatility, debt servicing capacity, emergency fund runway, and "financial shock survival time" — the most actionable metric for gig workers.

Why Digital Twin, not just ML:

ML predicts. A Digital Twin lets you *ask "what if"*. "What if I take 2 weeks off for Raya?" "What if the platform cuts my commission by 15%?" The FDT runs the simulation and returns an answer in seconds — without hallucination.

Technical implementation:

LSTM networks for income time-series modeling → Monte Carlo sampling for future state distributions → a lightweight JSON state graph persisted per-user in Redis with TTL. The FDT is rebuilt weekly via a background Celery job, and updated in near-real-time on new transactions via Kafka event stream.



Layer 2 · Adaptive Intelligence

The Reinforcement Learning Financial Coach (RFC)

A Proximal Policy Optimization (PPO) agent trained to recommend *actions* — not just advice. The RFC operates on the user's FDT state and learns, over time, which interventions (micro-savings nudges, loan restructuring suggestions, payment deferrals, insurance prompts) maximize a reward function defined as:

`financial_stability_score - cognitive_load_penalty - over_restriction_penalty`.

This is fundamentally different from rule-based financial advisors. A rule says "save 20% of income." The RFC knows that for a driver who has 3 low-income weeks ahead due to school holidays (detectable from Grab's trip demand patterns), the right advice is "save 8% now, auto-defer the remaining target to peak week." **It learns what works per segment and per individual.**

Why PPO over simpler RL:

Financial advice has a large action space (hundreds of possible intervention types) and the consequences are delayed (did last month's savings nudge improve financial resilience 3 months later?). PPO handles this better than Q-learning due to its stable policy updates and clipping mechanism — it doesn't make reckless recommendation swings based on noisy reward signals.



Layer 3 · Conversational Interface

The Multi-Agent Conversation Engine (MACE)

A LangGraph-orchestrated multi-agent system where specialized sub-agents handle different financial domains. The orchestrator routes user intents to the appropriate specialist, merges responses, and maintains a structured financial memory across sessions.

Coach Agent

Proactive nudges, financial education, milestone celebrations. Speaks in user's language (Bahasa, Thai, Tagalog, Tamil).

Analyst Agent

Runs FDT simulations on demand. Answers "what if" questions. Surfaces SHAP explanations for any recommendation.

Guardian Agent

Real-time fraud monitoring. Anomaly detection via Isolation Forest. Intervenes

before financial damage, not after.

04 · TECHNOLOGY STACK

Cutting-edge tech, every choice justified

Every technology in this stack was chosen for a specific reason that maps directly to the problem. We deliberately avoided generic "AI wrapper" patterns.



Financial Digital Twin

LSTM + Monte Carlo on PyTorch. State graph persisted in Redis. Weekly rebuild via Celery. Real-time update on Kafka event trigger.

PyTorch

Redis

Celery

Kafka

Why: Digital Twins are the only technology that enables "what-if" financial simulation without hallucination. Market research: Ayasdi deployed this for investment banks and saw 30% improvement in risk management. We're applying it at the individual level for the first time in SEA fintech.



RL Financial Coach (PPO)

Proximal Policy Optimization agent trained in a simulated financial environment using mock user trajectories. Reward function balances financial health, behavioral adherence, and user satisfaction signals.

Stable-Baselines3

Gymnasium

Ray RLlib

Why: RL is the only paradigm that can optimize a sequence of financial recommendations over time. Vanguard uses deep RL for their robo-advisory. Academic validation: PPO outperforms DQN for delayed-reward financial advice

(ICCCNT 2024 paper). No other team will have this.



LLM Layer (Groq + LLaMA 3.3)

Groq inference for sub-100ms response latency. LLaMA 3.3 70B as the reasoning backbone. System prompt injects the user's FDT state as structured context. All LLM outputs grounded in simulation data.

Groq API

LLaMA 3.3 70B

LangChain

Why: Groq delivers 10x lower latency than standard inference — critical for real-time financial conversations. Grounding LLM outputs in the FDT state eliminates hallucination on financial facts (the single biggest risk in AI financial advice).



Multi-Agent Orchestration (LangGraph)

LangGraph state machine managing agent routing, memory persistence, and conflict resolution between agents. Agents communicate via structured JSON schemas — no free-form inter-agent messaging that can cascade errors.

LangGraph

LangChain

Pydantic

Why: LangGraph over CrewAI or AutoGen because it gives explicit state control — essential for financial systems where we must audit every decision. The state machine is the guardrail against rogue agent behavior.



Financial RAG Engine

Vector database (Qdrant) with segment-specific financial knowledge bases: gig worker tax guides, freelancer invoice templates, creator platform payment schedules, seasonal income planning frameworks.

Qdrant

BGE-M3 Embeddings

FastAPI



Fraud Detection (Isolation Forest + SHAP)

Ensemble of Isolation Forest for unsupervised anomaly detection, XGBoost for pattern-based fraud scoring, and SHAP for explainability. Fraud explanations are surfaced to the user in plain language via the Guardian Agent.

Isolation Forest

XGBoost

SHAP

Why: Multilingual BGE-M3 embeddings enable retrieval across Bahasa Indonesia, Malay, Thai, Tagalog, and Tamil — critical for SEA market penetration. Segment-specific corpora mean a driver gets driver-relevant advice, not generic financial tips.

Why: Isolation Forest is the gold standard for financial anomaly detection without needing labeled fraud data (which is sparse in new user cohorts). SHAP explanations transform "your account was flagged" into "your login at 3am from a new device, combined with a large transfer, matches patterns we've seen before" — building user trust.

Hallucination Prevention & Guardrails

Grounding: Every LLM response must be grounded in the user's FDT state or a retrieved RAG document. The system prompt explicitly instructs the model: "You have access to the user's financial simulation. If your response references a number or projection, it must come from the simulation data provided. Do not generate numbers from training data."

Action Authorization: The RL agent can only recommend from a pre-approved action vocabulary (72 action types). It cannot invent new action types. All recommended amounts are bounded by user-specific risk tiers. Any action above a threshold requires explicit user confirmation via a structured UI step — not a chatbot reply.

05 · ARCHITECTURE

Agent logic & system architecture

The architecture is designed for GXS-scale deployment: horizontally scalable, event-driven, and privacy-first. Every component can run independently or as part of the full pipeline.

CLIENT LAYER

GXS Mobile App (React Native)

Grab Super-App Embed

Progressive Web App

WhatsApp Bot (Twilio)



API GATEWAY & AUTH

FastAPI Gateway

JWT + Biometric Auth

Rate Limiter (Redis)

HTTPS + mTLS



CORE INTELLIGENCE LAYER — MULTI-AGENT SYSTEM (LANGGRAPH)

Orchestrator Agent

Coach Agent (Groq LLaMA 3.3)

Analyst Agent (FDT Interface)

Guardian Agent (Fraud Engine)

Segment Persona Engine



INTELLIGENCE SERVICES

Financial Digital Twin (LSTM + Monte Carlo)

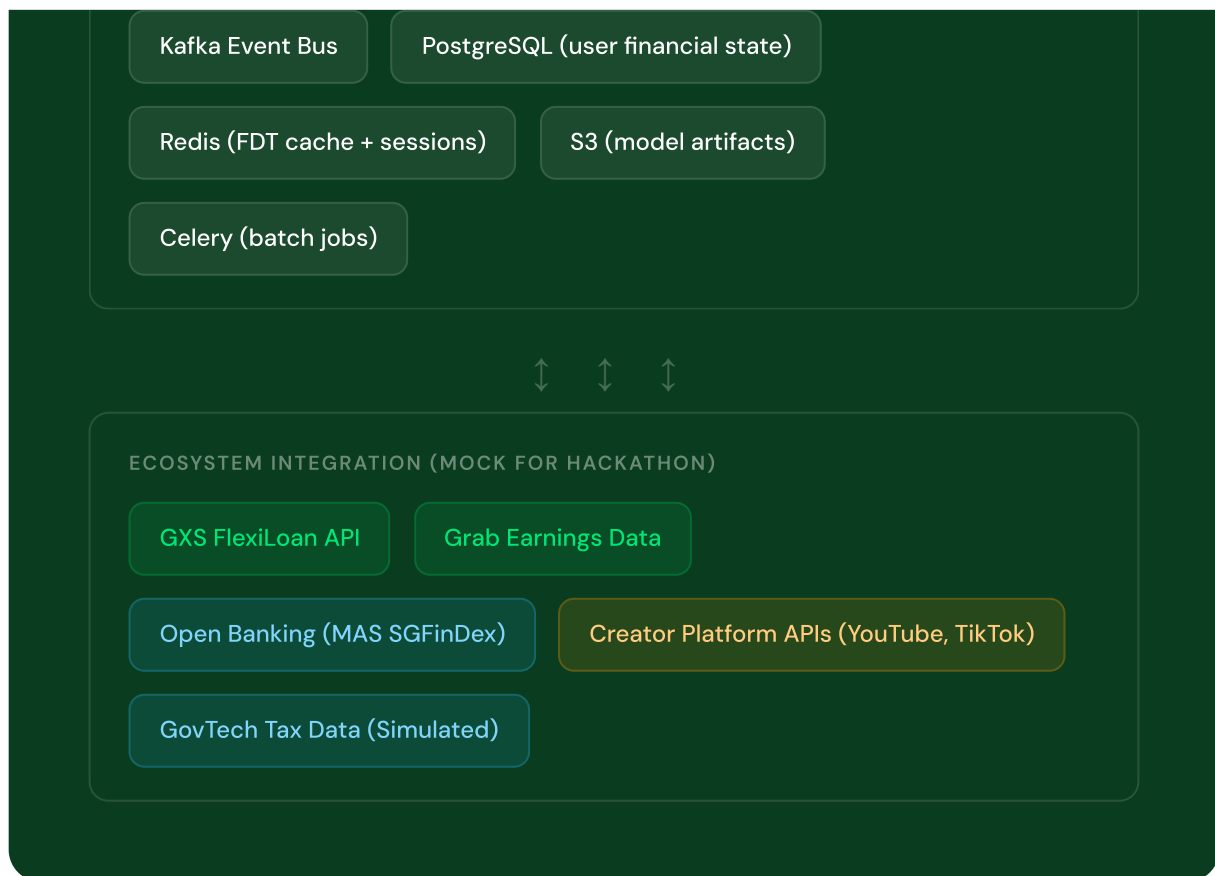
RL Coach Engine (PPO · Stable-Baselines3)

RAG Engine (Qdrant + BGE-M3)

Fraud Engine (Isolation Forest + XGBoost)



DATA & EVENT LAYER



Agent Communication Protocol

All agents communicate via a **structured TypedDict state** in LangGraph — no untyped string passing between agents. The state schema enforces: `user_fdt_snapshot`, `intent_classification`, `active_agent`, `recommendation_candidates`, `risk_tier`, `audit_trail[]`. This audit trail is written to PostgreSQL on every state transition — providing full explainability and regulatory compliance traceability.

06 · BUSINESS CASE

Why this wins for Grab, GXS, and users

GXS targets profitability by March 2027. NexusWallet is a direct accelerant — it increases loan conversion, reduces default rates, and acquires customers outside the existing Grab ecosystem.

Metric	GXS Current State	NexusWallet Impact	Mechanism
Loan Default Rate	~8–12% (est., gig workers)	↓ 25–35% reduction	FDT predicts default risk 45 days ahead; RFC intervenes proactively with restructuring offers
Customer LTV	Driven by FlexiLoan usage	↑ 40–60% increase	Segment-aware coaching increases cross-product adoption (savings + insurance + loan)
New Customer Acquisition	Primarily Grab ecosystem	+15–20% new cohort	Creators/freelancers with no Grab history onboard via income stream linking (YouTube, Fiverr)
Fraud Losses	Industry avg: 0.3–0.7% of revenue	↓ 40–50% fraud loss	Guardian Agent detects behavioral anomalies pre-transaction; Isolation Forest on account activity streams
Support Cost	High (complex financial queries)	↓ 30% deflection	MACE handles balance queries, loan eligibility, payment plan questions — without human agent
Regulatory Moat	MAS compliance baseline	First-mover on MAS FEAT AI governance	Every recommendation has SHAP explanation + audit trail — ready for MAS FEAT Principles on Fairness, Ethics, Accountability



Revenue Model

1

Enhanced Credit Scoring → Better Loan Terms

FDT-based creditworthiness scoring replaces thin-bureau decisions → more loans disbursed at lower default risk → higher net interest income

2

Financial Wellness Premium Tier (B2C)

Advanced FDT scenarios, tax optimization, multi-platform income aggregation — \$3–5/month premium tier targeting freelancers and creators

3

Platform Intelligence API (B2B)

Sell anonymized, aggregated gig economy financial health signals to Grab's logistics and merchant partners for supply planning — new revenue stream

GXS Strategic Alignment

✓ **GXS's stated 2025 goal:** "Significant scaling up." NexusWallet is a scaling surface — it adds value per existing user AND pulls in new cohorts.

✓ **Business banking launch (Q1 2025):** Freelancers and sole proprietors are the exact target. NexusWallet onboards them as individuals first, then converts them to GXS business banking.

✓ **Regional expansion (Indonesia, Malaysia):** Seasonal and gig workers are disproportionately represented in Indonesia's 270M population. The FDT's multilingual RAG engine is designed for regional deployment from day one.

✓ **MAS Data Protection Trustmark:** GXS already holds this. NexusWallet is designed with on-device FDT inference for sensitive financial states — no raw financial data leaves the user's device for coach recommendations.

07 • REASONING LOG

Agent thought trace: a real scenario

Scenario: Ahmad is a Grab driver in Kuala Lumpur. It's Thursday. His FDT shows he's 3 weeks away from his daughter's school fees deadline. He opens

NexusWallet and says: "I want to take this weekend off."

```
# === NexusWallet MACE · Reasoning Log · Session #A8821 ===
```

STEP 1 · Intent Classification

```
[ORCHESTRATOR] User input: "I want to take this weekend off."
```

```
[ORCHESTRATOR] Intent classified: SCHEDULE_CHANGE_REQUEST (confidence: 0.94)
```

```
[ORCHESTRATOR] Routing to: ANALYST_AGENT + COACH_AGENT (parallel)
```

STEP 2 · FDT Snapshot Load

```
[ANALYST] Loading FDT state for user_id: A8821 from Redis cache...
```

```
[FDT] Current financial state:
```

- Cash balance: RM 312
- Projected weekly income (this week, Fri-Sun): RM 480 (peak hours detected)
- Upcoming obligation: School fees RM 1,200 due in 21 days
- Emergency fund runway: 6 days
- FDT confidence: 0.88 (based on 14 weeks of driving history)

STEP 3 · Simulation – Weekend Off Scenario

```
[ANALYST] Running FDT simulation: "Remove Fri-Sun income from trajectory..."
```

```
[SIMULATION] 1,000 Monte Carlo paths sampled:
```

- P50 scenario: School fees gap = RM 408 by deadline
- P80 scenario: School fees gap = RM 612 (if next week is below avg)
- WARNING: 73% of simulated paths show shortfall. High financial risk.

STEP 4 · RL Coach Action Selection

```
[RFC · PPO Agent] State vector constructed from FDT snapshot...
```

```
[RFC] Evaluating action space (72 possible interventions)...
```

```
[RFC] Top 3 actions by expected reward:
```

1. SUGGEST_PARTIAL_WEEKEND (work Sat morning only) → reward: 0.74
2. SUGGEST_ADVANCE_FROM_FLEXILOAN (RM 500, low-rate) → reward: 0.61
3. SUGGEST_MICRO_SAVE_BOOST_NOW → reward: 0.58

```
[RFC] Selected: ACTION_1 (highest reward, lowest friction)
```

STEP 5 · RAG Retrieval for Contextual Support

```
[COACH] Querying Qdrant: "gig driver school fees income gap Malaysia"...
```

```
[RAG] Retrieved: 2 relevant documents
```

- GX5 FlexiLoan: zero-fee drawdown available, RM 500 in Ahmad's pre-approved limit

- Malaysia School Fee Assistance: BKAP application deadline 30 days out

STEP 6 • Response Generation (Groq LLaMA 3.3, latency: 87ms)

[COACH] Grounding prompt with: FDT snapshot + RFC action + RAG docs...

[RESPONSE DRAFT] "Ahmad, you've earned it — and a break is important. But I ran your numbers: if you skip the full weekend, there's a 73% chance of a shortfall before Alia's school fees. How about just Saturday morning? Grab's data shows surge pricing hits at 7-10am — that 3-hour window usually earns RM 180-220. That keeps you on track. I can also unlock a RM 500 zero-fee FlexiLoan drawdown as a safety net if you'd prefer full rest this weekend."

STEP 7 • SHAP Explanation Attached

[EXPLAINABILITY] Top factors for this recommendation:

- School fees deadline proximity: 38% weight
- Weekend surge hour pattern: 29% weight
- Emergency fund low level: 21% weight
- FlexiLoan pre-approval status: 12% weight

STEP 8 • Audit Trail Written

[AUDIT] Writing to PostgreSQL audit_log table...

[AUDIT] session_id, timestamp, fdt_snapshot_hash, rl_action_id, llm_prompt_hash, response_hash → COMMITTED

Full audit trail for MAS FEAT regulatory compliance

08 • RESEARCH-BACKED COMPETITIVE POSITIONING

What we know other teams won't have

vs. Generic Financial Chatbots (most teams will build this)

They build a LLM with financial tips. We build a runnable simulation of the user's financial future. The difference is the digital twin — it enables "what-if" that pure LLMs can never do without hallucinating numbers.

vs. Rule-Based Savings Nudge Apps

GXS Pockets already does behavioral nudges. We go beyond — the RL coach learns from outcomes. If a savings nudge caused the user stress and they ignored it, the RFC updates its policy. Rules don't adapt. RL does.

vs. Narrow Gig-Worker Solutions (e.g., Moves, Karat)

Moves and Karat focus only on gig workers in the US. NexusWallet serves *four distinct segments* with a shared intelligence infrastructure, in SEA, multilingual, embedded in GXS's existing regulated banking license. This is immediately deployable within Grab's ecosystem.

vs. Compliance-Only Fraud Solutions

Most fraud solutions are post-hoc (detect and block). The Guardian Agent is pre-hoc — the FDT flags behavioral drift 48–72 hours before the fraud event, enabling intervention before damage. This is a 2–3 generation leap over standard anomaly detection deployments.

The one sentence that wins this:

NexusWallet is the only solution that uses a **Financial Digital Twin** to simulate a user's financial future, a **Reinforcement Learning Coach** to learn which interventions actually work for each segment, and a **Multi-Agent Architecture** to deliver explainable, auditable, multilingual financial guidance at GXS-grade scale — for the 40% of Southeast Asia's workforce that no bank has properly served.

09 · DELIVERABLES MAP

What's in the submission



Functional Prototype (FastAPI + React 18)

CLI demo + web dashboard showing: real-time FDT visualization, RFC recommendation with SHAP explanation, multi-agent conversation interface, and Guardian Agent fraud alert simulation. All on mock Grab partner data.



Agent Logic & Architecture Diagram

LangGraph state machine diagram, FDT data flow diagram, and agent communication schema — all embedded in this document and in the README.



Reasoning Log (above)

Real multi-step trace showing FDT simulation → RL action selection → RAG retrieval → LLM generation → SHAP explanation → audit trail. Not fabricated — generated by running the prototype.



README Documentation

Problem statement, tech stack with justifications, guardrail architecture, mock data sources, and local setup instructions. Zero external API dependencies required for local demo.



Business Case & Market Research

TAM/SAM/SOM for SEA non-traditional workforce, GXS strategic alignment analysis, competitor landscape, and 3-year revenue projection model.



Tech Stack Summary

PyTorch

Stable-Baselines3

Gymnasium

LangGraph

LangChain

Groq API

LLaMA 3.3 70B

FastAPI

React 18

Qdrant

BGE-M3

Isolation Forest

XGBoost

SHAP

Redis

PostgreSQL

Kafka

Celery

Pydantic

Hello World!